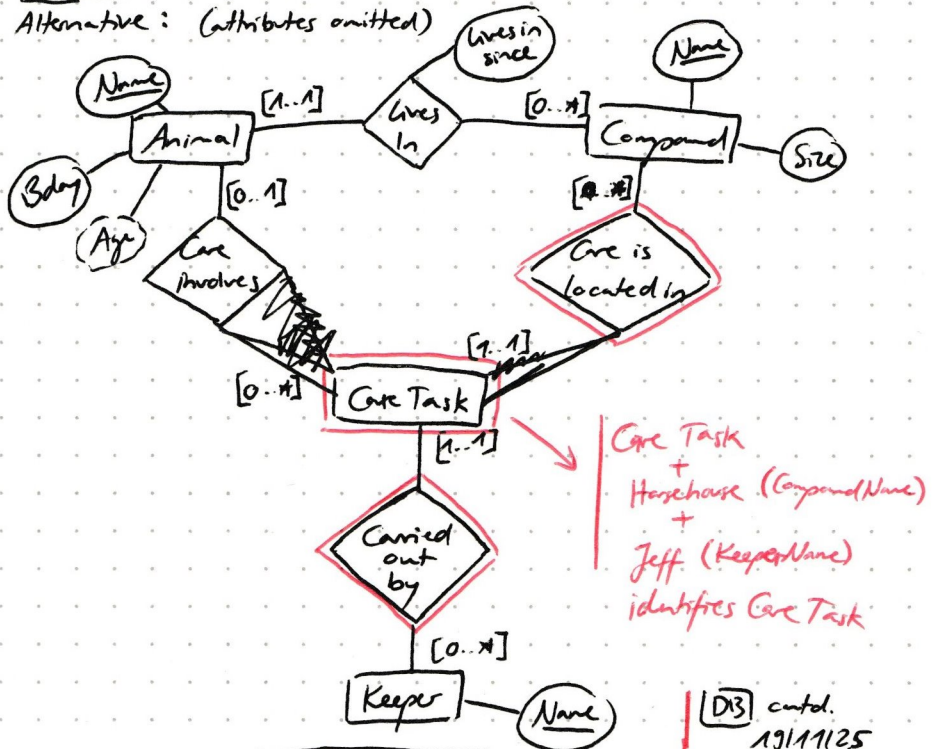


[DB] contd. 18/11/25
Alternative: (attributes omitted)

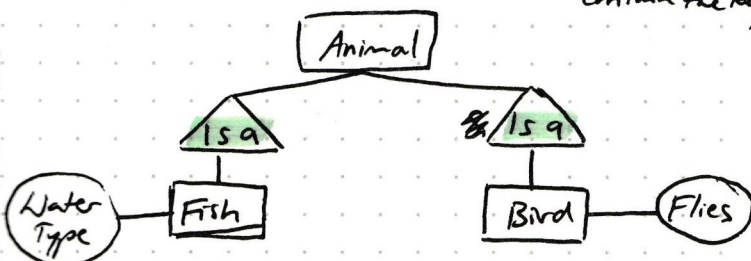


[DB] contd. 19/11/25

Specialization - is-a-relationship

→ cf. ~~inheritance~~ inheritance

- every root elem must contain the key

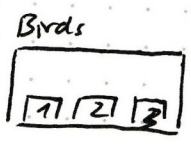
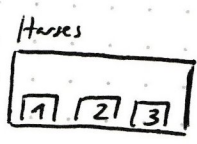
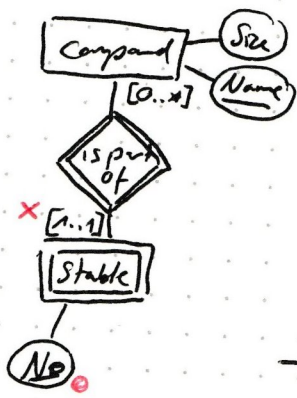


[DB] 19/11/25

Weak entity is an entity that has no primary key or just part of a primary key → marked with double line.

- completes its key using one or more **supporting relationships**

[D13] contd. 19/11/25

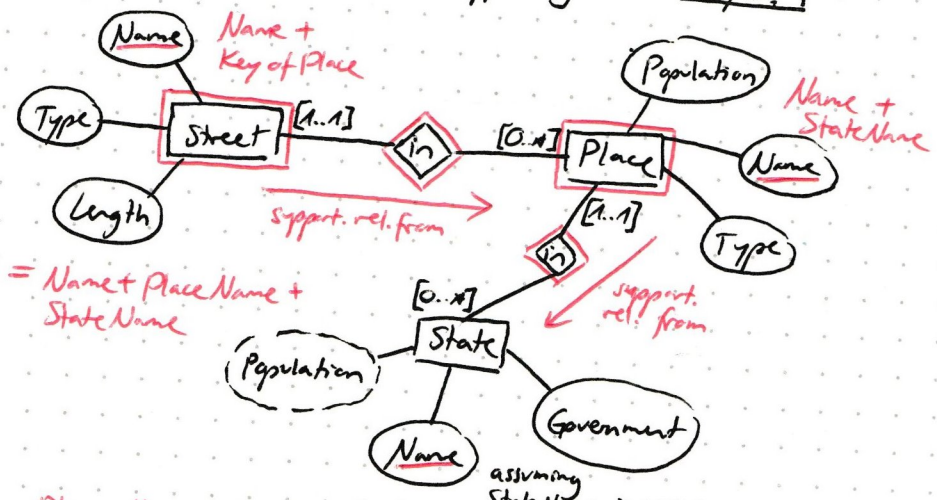


→ Stable No. alone does not ~~uniquely~~ uniquely identify the stable

→ Stable No. + Compound Name does uniquely identify a stable

- × ~~always~~ always the quantization for supporting relationships
- Part of the key but not the key

How to get weak entities and supporting relationships?



Place Name alone and Street Name alone can not uniquely identify them, as ~~any multiple states can have~~ there can be multiple streets/places in different places/states with the same name.

DB

19/11/25

$R = (A, B, C, D, E, F)$

note: Synthesis would do the same as the Decompose

$AB \rightarrow CDE$
 $D \rightarrow F$

a) D not superkey and F not prime

b) $R_1 = \{D\}^+ = \{D, F\}$ in BCNF

$R_2 = \{DABCE\}$ in BCNF

done!

$S = (VWXYZ)$

case was already applied
 $F = F_c$

$V \rightarrow WZ$
 $WZ \rightarrow VXY$
 $Y \rightarrow Z$

Keys = $\{V\}, \{WZ\}, \{WY\}$

$\{Y\}^+$ is no superkey but Z is prime $\rightarrow$ 3NF

$S_1 = \{Y\}^+ = \{Y, Z\}$

$S_2 = \{Y, V, W, X\}$

these got lost, not dependency preserving

stop, does not work

decomposition does not work and there is no other starting from since $Y \rightarrow Z$ is the only non-compliant FD.

formal notation:
 $\{Y\}^+ = \{Y, Z\}$
 $\Rightarrow S_1 = (Y, Z)$

Synthesis

~~$S_1 = (VWZ)$~~

~~$S_3 = (Y, Z)$~~

$S_2 = (WZVXY)$

$S_1 \subset S_2$
 $S_3 \subset S_2$ both are subsets of S_2

Problem: duplication of types

$\rightarrow$ not working, still in 3NF

$R = (ABCDEF)$

Synthesis

$A \rightarrow BC$
 $D \rightarrow EF$
 $\{$

Key = $\{A, D\}$

$R_1 \{ A \rightarrow B$
 $A \rightarrow C$
 $R_2 \{ D \rightarrow E$
 $D \rightarrow F$

$R_1 = (ABC) \rightarrow$ not lossless without R_3

$R_2 = (DEF)$

$R_3 = (AD)$ for lossless

$ABCD \rightarrow EF$

$\{ADF\}^+ = R \cup \emptyset$

[DB]

contd.
19/11/25

$$FD_{S_1} \equiv FD_{S_2}$$

$$S_1 = (APHRODITE)$$

$$R \rightarrow OD$$

$$Keys = \{ITER\}, \{ITEO\}$$

$$O \rightarrow AHPR$$

not in BCNF bc R and O not superkey

$$S_1 = \{R\}^+ = \{RODAHP\}$$

in BCNF

~~$S_1 = \{R\}^+ = \{RODAHP\}$~~

$$FD_{S_2} \equiv \text{Cover}(\{ \})$$

to be exact:

cannot be expressed as just $\{ \}$ bc of trivial FDs

$$S_2 = \{R|ITE\} \rightarrow \text{in BCNF}$$

Synthesis: $S_1 = (ROD)$

Decomp: less relation
less joins necessary

$$S_2 = (OAHPR)$$

$$\rightarrow \text{add } S_3 = (ITER) \text{ or } (ITEO)$$

Check BCNF: $\text{Cover}(FD) \rightarrow \text{Cover}(FD_{S_2}) = \{ \} = FD_{S_2}$

so the attributes in S_2

are superkey $\{R|ITE\}^+ = \{R|ITE\}$

[DB] contd.

19/11/25

Lecture Notes

Ex: What about Core Task?

- when actually implementing, it'll then make sense to add surrogate keys to identify the types eg. Keeper entity

Keeper ID* Name ← based on ER diagram
would be unique

Ex: Device ID

LT Type
R0 Place
B Building
1 Floor
18 Room
01 Device

Label: LTR0B11801

PBFRD $\rightarrow T$ in BCNF

